

CCSHAU Website — CI/CD Step-by-Step Guide

Repository: <https://github.com/SiyanainfoAhm/ccshau-website>

Production branch: `main`

Stack: Next.js · Supabase · Vercel · GitHub Actions

Generated for the CCSHAU monorepo (`apps/web`).

Overview

Step	Where	What happens
CI	GitHub Actions (<code>.github/workflows/ci.yml</code>)	lint → test → build
CD	Vercel	preview on PR, production on <code>main</code>
DB	Supabase	migrations are manual (not auto in CI yet)

Part A — One-time setup

1. GitHub repository access

- You need **push** access to `SiyanainfoAhm/ccshau-website` .
- Enable Actions: GitHub → **Settings** → **Actions** → **General** → allow workflows.

2. Connect Vercel to GitHub

In vercel.com:

- Add New Project** → import `ccshau-website`
- Root Directory:** `apps/web` (monorepo — required)
- Framework:** Next.js (auto-detected)
- Production branch:** `main`
- Enable **Preview Deployments** for pull requests

Vercel config file: `apps/web/vercel.json`

- Install from monorepo root
- Deploy region: `bom1` (Mumbai)

3. Vercel environment variables

Vercel → **Project** → **Settings** → **Environment Variables**

Required (Production + Preview):

Variable	Notes
<code>NEXT_PUBLIC_SUPABASE_URL</code>	e.g. <code>https://fvveqziyusjgqejowkfp.supabase.co</code>
<code>NEXT_PUBLIC_SUPABASE_ANON_KEY</code>	Public anon key
<code>SUPABASE_SERVICE_ROLE_KEY</code>	Server only — secret

NEXT_PUBLIC_SITE_URL	Production URL; preview can use Vercel URL
NEXT_PUBLIC_AZURE_STORAGE_ACCOUNT	Blob storage account
NEXT_PUBLIC_AZURE_STORAGE_CONTAINER	Container name
AZURE_STORAGE_CONNECTION_STRING	Server uploads — secret
CRON_SECRET	Production cron routes

Optional (when features enabled):

- POWER_AUTOMATE_EMAIL_URL , POWER_AUTOMATE_WEBHOOK_SECRET
- CAPTCHA_SECRET_KEY , NEXT_PUBLIC_CAPTCHA_SITE_KEY

Reference: `apps/web/.env.example` , `docs/phase-0/supabase-setup.md`

4. GitHub secrets (optional — backup workflow)

For `.github/workflows/daily-backup.yml` only:

- SUPABASE_ACCESS_TOKEN
- SUPABASE_DB_PASSWORD
- NEXT_PUBLIC_SUPABASE_URL
- SUPABASE_SERVICE_ROLE_KEY

Skip if you only need app CI/CD.

Part B — End-to-end release flow

Step 1 — Local pre-check

From repository root:

```
cd C:\Jatin\Projects\CCSHAU_Project
npm ci
npm run lint
npm test
npm run build
```

All four must pass before pushing. This mirrors GitHub Actions CI.

Optional (dev server must be running):

```
npm run dev
# new terminal:
npm run test:smoke
```

Step 2 — Create a feature branch

```
git checkout main
git pull origin main
```

```
git checkout -b feat/your-feature-name

# Stage and commit your changes
git add <files>
git commit -m "Describe your change"
git push -u origin feat/your-feature-name
```

Do not commit debug scripts, `.next` build output, or local probe artifacts unless intended.

Step 3 — Open a Pull Request

1. Go to GitHub → `ccshau-website`
2. Click **Compare & pull request**
3. Base: `main` ← Compare: your feature branch
4. Add title and description
5. Create PR

This triggers **CI (GitHub Actions)** and **CD preview (Vercel)** in parallel.

Step 4 — Watch CI (GitHub Actions)

GitHub → **Pull requests** → **your PR** → **Checks**

Job: **CI / lint-and-build**

```
✓ Install dependencies (npm ci)
✓ Lint
✓ Unit tests (227 Vitest cases)
✓ Build
```

If CI fails: read the log, fix locally, commit, push — CI re-runs automatically.

Actions URL: <https://github.com/SiyanainfoAhm/ccshau-website/actions>

Step 5 — Watch CD preview (Vercel)

On the PR, the Vercel bot posts a **Preview URL**, e.g.:

```
https://ccshau-website-xxxxx.vercel.app
```

Or check Vercel dashboard → **Deployments**.

Verify on preview:

- Homepage loads
- `/admin/login` loads
- A college microsite page loads
- Images/uploads work (Azure env must be set for Preview)

Step 6 — Review and merge

When CI and preview look good:

1. Code review (self or teammate)
2. **Merge pull request** → confirm merge

3. Optionally delete the feature branch

Step 7 — Production deploy (automatic)

Merge to `main` triggers:

System	Action
GitHub Actions	CI runs again on <code>main</code>
Vercel	Production deploy starts

Check Vercel → **Deployments** → status **Ready**.

Smoke-test production:

- Homepage
- `/admin/login`
- One admin/CMS action (if you have access)

Step 8 — Database migrations (if applicable)

If the release includes Supabase migrations under `supabase/migrations/` :

```
npx supabase link --project-ref fvveqziyusjgqejowkfp
npx supabase db push
```

Run **after merge** — not automated in CI today.

Part C — Post-deploy verification

Check	How
CI green on <code>main</code>	GitHub Actions tab
Vercel production Ready	Vercel dashboard
Site live	Open production domain
Health API	<code>GET /api/health → { "status": "ok" }</code>
Admin login	Visit <code>/admin/login</code>
Images/uploads	Upload in admin; view on public page

Part D — Troubleshooting

Problem	Likely cause	Fix
CI fails on <code>npm test</code>	Failing unit test	Run <code>npm test</code> locally; fix; push
CI fails on <code>build</code>	TypeScript or build error	Run <code>npm run build</code> locally
Vercel build fails	Wrong root directory	Set root to <code>apps/web</code>

Preview OK, production broken	Missing Production env vars	Add vars for Production in Vercel
Blank images on preview	Azure vars missing in Preview	Add Azure env to Preview scope
No Vercel link on PR	Vercel not linked to repo	Re-import / reconnect project

Part E — CI workflow reference

File: `.github/workflows/ci.yml`

Triggers: push and pull_request to `main` or `develop`

Steps:

1. Checkout code
2. Setup Node.js 20 (npm cache)
3. `npm ci`
4. `npm run lint`
5. `npm test`
6. `npm run build` (placeholder Supabase env vars in CI)

Part F — Recommended first E2E run

1. Confirm Vercel project exists with root `apps/web`
2. Set Production + Preview environment variables
3. Run locally: `npm ci && npm run lint && npm test && npm run build`
4. Push a small feature branch
5. Open PR → wait for green CI + Vercel preview
6. Test preview URL
7. Merge to `main`
8. Confirm production deployment

Related files

File	Purpose
<code>.github/workflows/ci.yml</code>	Main CI pipeline
<code>.github/workflows/daily-backup.yml</code>	Daily Supabase backup (ops)
<code>apps/web/vercel.json</code>	Vercel build config
<code>Documents/ci-cd-pipeline.mmd</code>	Visual flowchart source
<code>Documents/ci-cd-pipeline.png</code>	Visual flowchart image
<code>scripts/render-ci-cd-pdf.mjs</code>	Regenerate flowchart PDF